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Rotation of the Oman ophiolite and initial location of the ridge in the hotspot reference frame

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Abstract

The volcanic units of the Oman ophiolite were paleomagnetically studied in order to better constrain the large rotation of the nappe during emplacement suggested by previous paleomagnetic results. The volcanic sequence, going from pillow basalts extruded at the mid-oceanic ridge up to submarine lavas extruded during the oceanic thrusting, before the end of the obduction, displays components of magnetization which were acquired at various stages during the movement of the nappe on the sea floor. Paleomagnetic Euler Pole analysis shows that the movement can be modeled by a single, large rotation of about 145–150° around an Euler pole located close to the nappe (less than 200 km away). Combining plate tectonic concepts, a mantle-anchored hotspot reference frame and paleomagnetic data, we were able to determine the initial paleolatitude (around 10°N), the initial paleolongitude (around 45°E) as well as the paleorientation (NNW–SSE to N–S) of the ridge in hotspot coordinates. The linear velocity of the nappe between the period of accretion at the ridge and the end of the obduction onto the Arabian peninsula would have been slightly less than 2 cm per year.

1. Introduction

The Oman nappe is the largest ophiolitic complex known in the world, with a crescent-shaped extension of over 500 km in length and 50–100 km in width. The ophiolitic nappe itself is divided into structural massifs, each several tens of kilometers long (Fig. 1). The nappe was obducted on the Arabian platform at the end of the Cretaceous. It has not been disturbed by any subsequent collision but it underwent a slight tilt during a post-Cretaceous tectonic event. For this reason, the ophiolitic section is complete, from the metamorphic sole resting on sedimentary nappes, through peridotites, gabbros and the sheeted dike complex, to the volcanic units interbedded with pelagic oceanic sediments. It is one of the best studied ophiolitic complex and many special volumes have been published ("Oman ophiolites", J. Geophys. Res., vol. 86, 1981; "The ophiolites of Northern Oman", Geol. Soc. Mem., vol. 11, 1986; "The ophiolite of Oman", Tectonophysics, vol. 151, 1988).

Large rotations during intraoceanic thrusting have been proposed for the Oman ophiolitic nappe (Thomas et al., 1988; Thomas, 1991) based on paleomagnetic results from gabbros (Luyendyk and Day, 1982; Shelton, 1984; Thomas, 1991),

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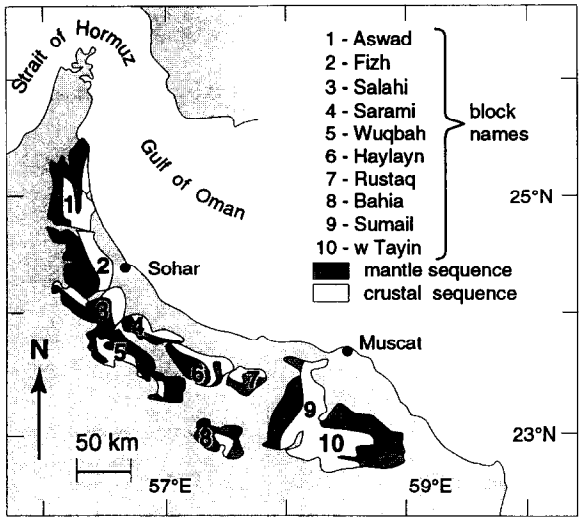


Fig. 1. Outlines of the Oman ophiolite after Ernewein et al. (1988).

dikes (Luyendyk et al., 1982; Shelton, 1984) and sediments (Thomas, 1991). The chronology between the different groups of magnetic directions and their polarity remain, however, questionable. The sense of rotation of the nappe is, therefore, still controversial (Thomas et al., 1988; Thomas, 1991).

Volcanic units of the sequence had not been paleomagnetically studied before, but were selected for various reasons. Firstly, the reliability of volcanic rocks as paleomagnetic recorders might lead to better results than from intrusive or sedimentary units. Secondly, the presence of interbedded and overlying sediments allowed determination of the paleohorizontal and provided good age control. Finally, some massifs comprise a sequence of volcanic rocks from pillow basalts extruded at the mid-oceanic ridge, before the detachment of the nappe, up to lavas extruded before the end of the obduction (see Tectonophysics, special issue, 1988). Thus, we can hope to obtain successive discrete records of the nappe movements on the sea floor. The Salahi massif, located in the northern part of the Oman mountain (Fig. 1), was chosen for the paleomagnetic analysis because of its complete and particularly well developed volcanic sequence.

2. Geological setting and sampling

Ernewein et al. (1988) distinguish three episods of submarine volcanism on petrologic and geochemical grounds. A first episode (V1) of ridge-axis volcanism, with pillow basalts and some massive flows that directly overly the dike complex. A second volcanic episode (V2) comprises two lava facies with different degrees of partial fusion. These V2 lava flows clearly postdated the V1 episodes. However, absence of a continuous sedimentary layer between the two episodes indicates no substantial interruption in volcanic activity between the two episodes. Biostratigraphic ages determined from radiolarian faunas present in the pelagic sediments interbedded with the volcanic sequence indicate that the first two volcanic episodes occurred within a very short time interval. According to Ernewein et al. (1988), the V1 episode took place from late Albian to early Cenomanian (approx. 100 Ma) at the mid-oceanic ridge and the V2 episode occurred shortly afterwards at the start of oceanic thrusting during Cenomanian to Turonian times (around 90 Ma). The third volcanic episode (V3) is characterised by thick columnar flows with doleritic texture. A thick pelagic sedimentary layer (up to 15 m) is observed between the V2 and V3 units. This V3 episode is interpreted as due to intraplate seamount volcanism, produced some 15–20 m.y. after the beginning of oceanic thrusting, probably during Santonian to Campanian times (around 70 Ma), but preceding the end of the obduction of the ophiolite onto the Arabian continental margin (Ernewein et al., 1988). Hydrothermal circulations, contemporaneous with the various phases of volcanic activity, appear to have induced some recrystallization in the lavas and interbedded sediments (Karpoff et al., 1988; Pflumio, 1988).

Paleomagnetic samples were collected along four sections (Fig. 2; Table 1) in the volcanic units of the Salahi massif. This massif is delimited by two main fault zones: the Wadi Jizzi fault zone in the north and the Wadi Ahin fault zone in the south. The volcanic sequence of this massif is particularly well preserved with an average thickness on the order of 2.2 km. Section S1 covers the V1 unit (from the contact with the sheeted dike



Fig. 2. Geologic map of the sampling area from Pflumio (1988) and location of the paleomagnetic sections (*S1–S4*). *1* = dike complex; *2* = V1 unit; *3* = V2 unit; *4* = V3 unit; *5* = intrusive gabbros; *6* = plagiogranites; *7* = metalliferous sediments; *8* = fault contact; *9* = ancient gravel terraces; *10* = wadi gravels and present-day terraces.

complex up to the contact with the V2 unit) and the bottom part of the V2 unit. Section S2 covers the lower part of the V1 unit up to the basal V2 lava flow. The S3 section starts in the upper part of V2 and finishes in the V3 lavas while section S4 lies entirely in the basal part of the V3 unit. Samples were drilled in the field using a gasoline-powered core drill and oriented using both magnetic and sun compasses. Each sample was cut in three to seven standard size specimens (2.5 cm in diameter and 2.2 cm in height).

3. Paleomagnetic and magnetic analysis

Remanent magnetization was measured using a CTF cryogenic magnetometer. Magnetic viscosity of the samples was estimated by the Thellier's procedure (Thellier and Thellier, 1944). The viscosity index is very low: less than 5% for 90% of the samples with a maximum value of 9%. Thus, the VRM acquired in the laboratory is small. One hundred and thirty three specimens were demagnetized either thermally or by alternating fields (AF). Stepwise thermal treatments up to 600°C were applied to half of the specimens with 50°C steps. The magnetic susceptibility at room temperature k_0 was measured after each heating step using a Bartington bridge. AF demagnetization up to 150 mT was performed on the rest of the specimens with 5, 10 or 20 mT steps. Paleomagnetic analysis of these specimens is straightforward. After removal, in most cases, of a present-day overprint, stable components of magnetization were determined through principal component analysis (Kirschvink, 1980). These final directions are presented in stratigraphic coordinates thus corrected for local tilting. Tilt corrections (Table 1) correspond to the average tilt of the interbedded sediments, which is taken as the best estimate of the local paleohorizontal. Hysteresis loops up to 1 Tesla were determined for 32 specimens from V1, V2 and V3 units using the equipment of the Laboratoire de Géomagnétisme du Parc Saint Maur. Thermal variation of low-field susceptibility was studied using a Bartington bridge with furnace in order to determine the Curie points of some samples (V1 pillow basalts).

3.1. Unit V1

With the exception of one possible sill, all V1 samples were drilled from pillow basalts. Magnetic properties of fresh pillow basalts from present-day accretionnary plate boundaries have been intensively studied, both for slow- and fast-spreading ridges (Prévot et al., 1979; Smith-Daignières, 1984). It is clear that, on the basis of major differences in their magnetic properties compared with modern pillow basalts, the magnetic mineralogy of the V1 pillow basalts has been strongly modified, subsequent to their eruption on the sea floor: (1) their intensity of magnetization J_r is 10 to 15 times weaker; (2) their

Table 1
Sampling informations

Section	Site coordinates		Tilt correction (DDD)		Sample no.	Volcanic units
	Lat.	Long.	Azimuth	Dip		
S1	24.15°N	56.52°E	116°	28°	44	V1–V2
S2	24.14°N	56.53°E	134°	23°	42	V1–V2
S3	24.19°N	56.55°E	73°	25°	37	V2–V3
S4	24.16°N	56.55°E	78°	30°	10	V3

DDD = Down dip direction

Koenigsberger ratio Q_n is smaller by two orders of magnitude; and (3) they show the presence of a nearly Ti-free titanomagnetite phase (near-magnetite phase), in addition to the low-temperature phase which is typical of submarine basalts. For the V1 pillow basalts, the latter mineral is thermally unstable even under vacuum, which suggests a highly oxidized titanomaghemite.

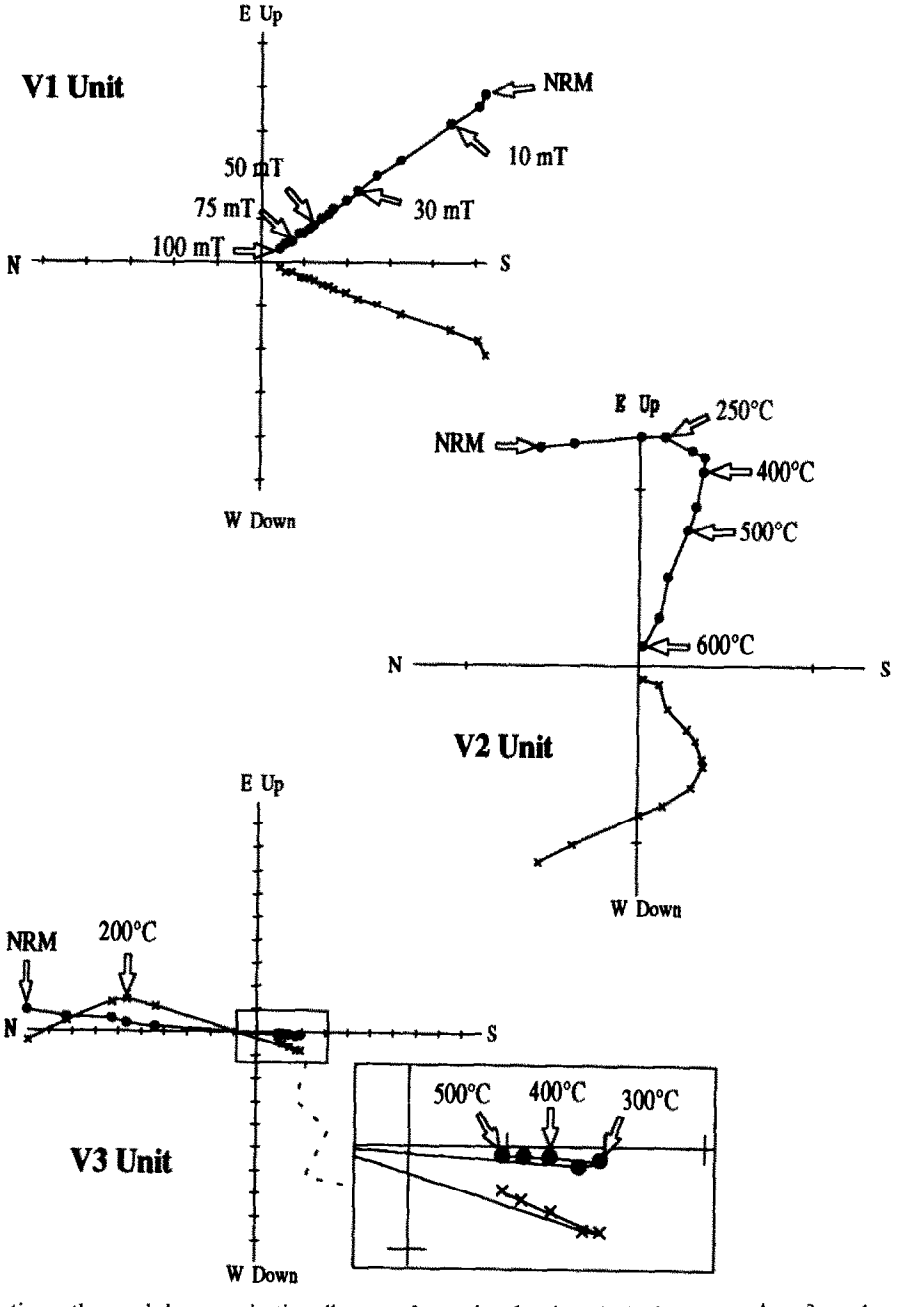


Fig. 3. Representative orthogonal demagnetization diagrams for each volcanic unit. Units are 10^{-4} Am^2 kg^{-1}. Dots and crosses represent the horizontal and vertical planes, respectively.

It is believed that maghemitization of titanomagnetite in pillow basalts does not alter their direction of remanence (Prévot et al., 1981; Özdemir and Dunlop, 1985; Brown and O'Reilly, 1988). Thus the low-temperature component of magnetization of the V1 pillow basalts is expected to have preserved the direction of the original TRM. Maghemitization develops very rapidly at the ridge axis: very young pillow basalts from the rift valley of the Mid-Atlantic Ridge are already significantly maghemitized (Prévot et al., 1983). We suggest that the near-magnetite phase results from partial inversion of titanomaghemite, as a result of hydrothermal processes. This should result in a CRM carried by near-magnetite phases and directed either along the ambient field, or possibly intermediate in direction between the field and the TRM directions, as observed during multiphase oxidation of magnetite (Heider and Dunlop, 1987).

The present-day field overprint is minor and easily removed from most V1 samples. This leaves one very well defined component of magnetization (see a typical orthogonal diagram on Fig. 3). Both the low and high Curie-point phases yield generally the same direction, suggesting that inversion occurred shortly after eruption. All V1 directions plot in the southeastern quadrant of the equal-area projection (Fig. 4) but surprisingly with a much wider dispersion in declination than in inclination. This general trend is highlighted by the ellipse (Fig. 4) which covers 85% of the results. Volcanic rocks usually acquire their magnetization rapidly while cooling. Therefore, they would record secular variation with a fairly large but somewhat isotropic dispersion of primary directions. The anisotropic dispersion observed for the V1 samples seems to indicate that paleosecular variation is not, in this case, the only dispersion cause. Results from section S1 show a statistical decrease in declination while going up section. This is not obvious for the S2 section; although, a sill intruding the V1 sequence there provides also lower declination. Furthermore, some samples from section S2 showed two components of magnetization with the same inclination but a slightly different declination. The first component was defined up to 300–350°C while

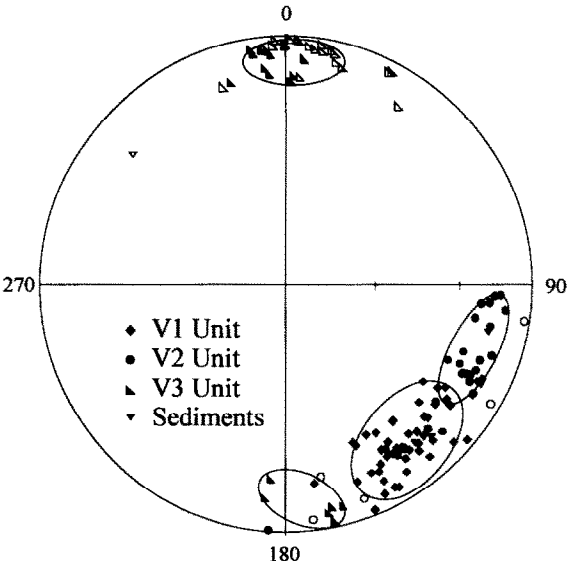


Fig. 4. Individual paleomagnetic directions obtained from the volcanic sequence and interbedded sediments (Thomas, 1991) presented on an equal area projection. Solid and open symbols represent positive and negative inclinations, respectively. The ellipses cover 85% of the individual results obtained for each volcanic unit. Both polarity V3 components of magnetization are presented on this figure.

the second with a lower declination was broken down at higher temperatures. The total of these observations suggests a temporal trend in declination.

3.2. Unit V2

In contrast to V1, the sampled V2 outcrops are made up mainly of thin flows (usually a few meters thick), possibly with some interstratified sills. Sea-floor alteration appears to have been much more intense than for V1. Present oceanic crust also shows flows in addition to pillow basalts and numerous studies of D.S.D.P. doleritic basalts of various ages indicate that their magnetization is carried by large homogeneous titanomagnetite grains with a low Curie point (100–300°C according to Smith-Daignières, 1984). Judging from the thermal demagnetization of NRM, the magnetic mineralogy of unit V2 seems to be more complex, often with both a low- (around 300°C) and a high- (500–580°C) temperature component of magneti-

zation whose relative magnitudes can be different from sample to sample. As for V1, this suggests some post-eruptive alteration of magnetic minerals, possibly also as the result of hydrothermal activity. Thermal or alternating field demagnetizations show a sometimes predominant recent overprint that is larger than for the V1 pillow basalts (see a typical orthogonal plot on Fig. 3). Nevertheless, after removal of this overprint, one stable high-temperature component of magnetization can be defined for most of the samples. Most of the directions (90%) have declinations between 90 and 130° (Fig. 4). Here again, the dispersion of directions is strongly anisotropic. The ellips, which contains 85% of the directions, slightly overlaps the V1 ellips (Fig. 4). The V2 samples present no direct evidence for a time control on the directional trend. A minor group of directions (10% of the samples) shows declinations around 180°. These directions, observed mainly in one flow located at the more altered upper part of section S1, are close to the directions found in the V3 unit and likely represent some later overprint.

3.3. Unit V3

Three very thick flows (several tens of meters) were sampled. The two flows from section S3 are separated by a thin sedimentary layer. At section S4, the lower part of the flow has pillow structures. The J_{rs}/J_s ratios range between 0.03 and 0.09, indicating the presence of multidomain magnetic grains. Thermal demagnetization of NRM reveals that between 65 and 90% of the remanence is unblocked below 300°C. Keeping in mind that the magnetic viscosity is low, we believe these low unblocking temperature components correspond to a primary thermoremanence carried by a low Curie-point titanomagnetite (or titanomaghemite), as observed in doleritic basalts from the present-day sea floor. The corresponding directions of magnetization are well defined (see a typical orthogonal diagram on Fig. 3) and well clustered within each flow, with a declination close to 0° (Fig. 4).

A distinctive feature of unit V3 is the presence of a second component of magnetization in most samples (Fig. 3). This is a high-temperature component, generally detected between 500 and 580°C, quite small in magnitude, and with a direction that is approximately reversed with respect to the low-temperature component (declinations around 180°, Fig. 4). The samples from the pillow basalt part of section S4 only do not show the reversed component. This reversed component can be interpreted as an overprint acquired during a later reversed period as a result of magnetite crystallization. Mineralogical studies are needed to determine the origin of the magnetite.

3.4. Conclusions

Most of the characteristic directions of remanence from units V1 and V2 are carried, at least in part, by titanium-poor titanomagnetite which probably crystallized as a result of mild heating due to hydrothermal activity. For the V1 unit, the general similarity in direction between the low- and high-temperature remanences suggests that this recrystallization occurred very shortly after eruption. This is also supported by the fact that the V1 and V2 directions are statistically significantly different (almost no overlapping in the ellips of confidence, Fig. 4 and Table 2) but very close to each other. A few blocks of sediments (umber) from the Salahi massif were studied by Thomas (1991). The directions belong to the same southeasterly directional trend as the V2 results except for one massif which gave a nearly antipo-

Table 2
Mean paleomagnetic directions and poles associated with the volcanic units of the Oman ophiolite

Unit	N	Inc.	Dec.	k	α_{95}	Plat.	Plong.	K	A_{95}
V1	35	20.2	145.8	80	2.7	−41.8	104.4	116	2.3
V2	18	17.6	106.5	51	4.9	−11.0	131.2	63	4.4
V3	27	1.3	1.9	58	3.7	66.4	231.7	104	2.7

N = number of specimens used in the mean calculation; Inc., Dec. = inclination, declination, respectively, of the mean direction of magnetization; Plat., Plong. = latitude, longitude, respectively, of the paleomagnetic pole associated with the mean direction of magnetization; (k, α_{95})–(K, A_{95}) = Fisher statistical parameters associated, respectively, with the mean direction of magnetization and with the paleomagnetic pole.

dal direction (Fig. 4). No examples of opposite polarity directions were found within either the V1 or the V2 units. Both V1 and V2 units were emplaced during the Cretaceous Superchron (long normal polarity interval), which lasted from the early Aptian (around 114 Ma, Odin and Odin, 1990) to the late Santonian (around 83 Ma, Odin and Odin, 1990), we consider their directions therefore to be of normal polarity. The V3 unit was emplaced during Santonian to Campanian times, therefore its primary magnetization can be as well normal than reversed. Furthermore the two polarities are found in this unit, so there is no paleomagnetic argument for deciding which polarity is primary. As long as the northerly magnetized V3 components of magnetization belong to the same directional trend as the V1 and V2 results, they are interpreted as normal. Thus we suggest that, despite evident remagnetization events, the set of directions from V1 to V3 is chronologically ordered and documents a progressive evolution of direction with time (Fig. 4).

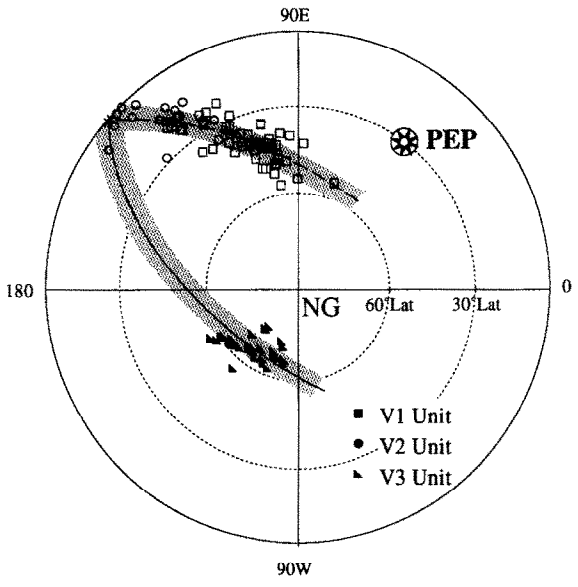


Fig. 5. Paleomagnetic Euler Pole PEP of rotation (star) and small circle with standard deviation (shaded area) which best fits the Virtual Geomagnetic Poles calculated from the individual components of magnetization defined for the volcanic units. Solid (open) symbols = north (south) latitudes. Polar equal area projection with *NG* as the north geographic pole.

4. Geodynamic implication of paleomagnetic data

The individual Virtual Geomagnetic Poles (VGP) corresponding to the characteristic directions of magnetization defined for the V1, V2 and V3 sequences (Fig. 5) can be fitted to a small circle with a radius of $84 \pm 4°$ and centered, in actual geographic coordinates, on a Paleomagnetic Euler Pole (PEP, Gordon, 1987) located at 29°N, 55°E (Table 3). Note that a calculation based solely on the V1 and V2 results gives a PEP only 10° apart from the former one (Table 3). The circular distribution of VGPs indicates

Table 3
Euler parameters of rotation

Reference system	*N*	PEP	d*x*/d*y*	$AD_{VGP} \pm$ s.d.	AD_S
Actual geographic	99 (V1 + V2 + V3)	29°N, 55°E	4.3/1.3	$84.5 \pm 4.1°$	
	72 (V1 + V2)	34°N, 44°E	20.8/3.8	$94.3 \pm 4.0°$	
	3 (means)	29°N, 53°E		85.1°	
Hotspot (not corr.)	99 (V1 + V2 + V3)	13°N, 40°E	4.3/1.3	$84.9 \pm 4.1°$	5.1°
	3 (means)	13°N, 39°E		85.1°	5.6°
Hotspot (corr.)	3 (means)	9°N, 44°E		82.2°	1.9°

PEP = Paleomagnetic Euler Pole (latitude, longitude); d*x*/d*y* = semi-axis of the ellipse of confidence around PEP; AD_{VGP} = angular distance between the Euler pole and the small circle which best fits the individual VGPS or the paleomagnetic poles distribution; s.d. = standard deviation; AD_S = angular distance at which the center part of the Salahi massif is rotating around the Euler pole; (not corr.), (corr.) = paleomagnetic poles not corrected, corrected, respectively, for TPW.

that the components of magnetization were acquired while the Oman nappe was rotating and that the overall movement of the Salahi massif can be described by a single clockwise rotation on the order of 145–150° between approximately 100 and 70 Ma. The VGPs obtained from the Salahi massif can be taken to be representative for the entire ophiolitic nappe only if no differential rotation did occur between the different massifs forming the ophiolite. This assumption, which still has to be confirmed, is supported by previous paleomagnetic results (see the compilation of results made by Thomas, 1991) and will be accepted for the rest of the analysis.

We also try to estimate the most significant paleomagnetic poles associated with the time of emplacement of the different units (Table 2). The low-temperature components of magnetization of the V1 unit, which are expected to have preserved the direction of the original TRM, were used to calculated the V1 paleomagnetic pole. For the V2 unit, we used all components of magnetization except those corresponding to the V3 remagnetization. Finally, for the V3 unit, we only considered the normaly magnetized components of magnetization supposed to be of primary origin. The corresponding Euler pole (29°N, 53°E, Table 3) is similar to the one obtained using individual VGPs (29°N, 55°E). The mean V3 paleomagnetic pole is not significantly different from the African Late Cretaceous paleopoles. The agreement becomes even better if we correct the V3 pole for the post-Miocene opening of the Red Sea (Fig. 6). Thus, the location of the Oman nappe with respect to the Arabian platform did not noticeably change between the emplacement of V3 (at ca. 70 Ma) and the Present. Using standard paleomagnetic procedure, we can then estimate the paleolatitude of the ridge, where V1 was erupted at 100 Ma, to be around 11°N ± 1°. When V2 was emplaced, shortly afterwards the start of the oceanic thrusting, the paleolatitude would have been around 9°N ± 3° and the nappe was almost equatorial, around 1°N ± 2° when V3 was magnetized. Paleomagnetism can also give information about the paleo-orientation of this portion of oceanic crust but cannot estimate the paleolongitude.

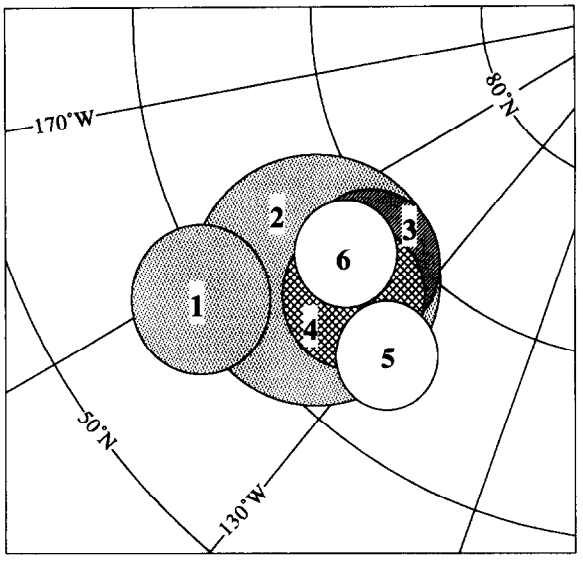


Fig. 6. Comparison between the paleomagnetic poles associated with the V3 unit (before and after closure of the Red Sea) and Late Cretaceous African data. *1*, *2* = Andriamirado, 1971; *3* = Storedvedt, 1979; *4* = Besse and Courtillot, 1991 (individual pole defined by small circles intersection); *5* = V3 unit's paleomagnetic pole; *6* = V3 unit's paleomagnetic pole corrected for the post-Miocene opening of the Red Sea.

In order to overcome this difficulty, we retrace the movement of the ophiolitic nappe in the hotspot reference frame. Hotspots appear to have rather slow relative motion in the mantle, and taken together they form a convenient reference system roughly fixed with respect to the mantle (Morgan, 1983; Duncan and Richards, 1991). We defined earlier that the location of the Oman nappe with respect to the Arabian platform did not noticeably change between the emplacement of V3 (ca. 70 Ma) and the Present. Therefore the rotation parameters which relocate Africa in its Late Cretaceous position in the hotspot reference frame (Morgan, 1983) can be applied, after closure of the Red Sea, to the Arabian platform and the ophiolitic nappe. Application of this rotation relocates the continental mass under consideration (the Arabian peninsula and the ophiolitic nappe) as well as the VGPs to their 70 Ma configuration in the hotspot reference frame. We can then determine the Euler pole associated with the clockwise rotation of the nappe in this reference frame (Fig. 7). This can be donc using

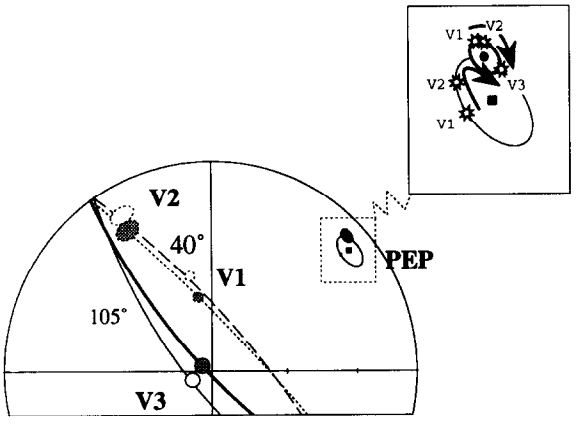


Fig. 7. Paleomagnetic Euler Poles and small circles which best fit the paleomagnetic poles associated with the volcanic units in the hotspot reference frame. Also presented on this figure is the small circle along which the center point of the Salahi massif (*V1–V3* stars in the inset) is rotating. The square, thin lines and empty ellipses correspond, respectively, to the Euler pole, the small circles and the ellipses of 95% confidence around the paleomagnetic poles calculated without correcting the paleomagnetic poles for true polar wander. The same parameters calculated after correction of the paleomagnetic poles for true polar wander are similarly represented by the circle, thick lines and shaded ellipses. Polar equal area projection.

either the individual or the mean VGPs (Table 3). Knowing the Euler pole and the angle of rotation, and the 70 Ma position of the Oman nappe, we can then describe the rotation of the nappe in the hotspot reference frame (Inset on Fig. 7). The error on the PEP location is on the order of 4° and the distance between the PEP and the Oman nappe is on the same order. Therefore, this calculation does not exclude the possibility of a PEP location next to or on the nappe itself. We can approximate the paleoposition of the spreading ridge at ca. 100 Ma with a paleolatitude around 10–20°N and a paleolongitude around 40–45°E. Taking a 5° circular track along which the nappe has been rotating, we can estimate the linear velocity of the nappe to be around 5 cm/yr.

The paleolatitudes defined for the Oman ophiolite in the hotspot reference frame are around 8°N at 70 Ma, 16°N at 90 Ma and 18°N at 100 Ma. Comparison of these paleolatitudes with the paleolatitudes deduced in the paleomagnetic reference frame shows a systematic shift on the order of 7°. This shift can be caused by non-dipole field influence, inacccurate reconstructions of the plate motion relative to the hotspots or by motion of the entire Earth relative to its axis of rotation. Several studies (Livermore et al., 1984; Andrews, 1985; Gordon, 1987; Besse and Courtillot, 1991; Duncan and Richards, 1991) indicate that True Polar Wander (TPW, i.e., displacement of the mantle with respect to the geomagnetic axis) did occur between 100 and 70 Ma, during emplacement of the ophiolitic nappe. If this is the case, the rotation, we just defined, is not an Euler rotation but result from the combination of two distinct movements: the movement of the mantle relative to the Earth rotation axis and the movement of the nappe on the sea floor. In order to define solely the movement of the nappe, we shall remove the influence of TPW from the VGPs. If we considered that the movement of the mantle with respect to the geomagnetic axis can be modelled by a Euler rotation, we can rotate the paleomagnetic coordinate system in order to bring the Master Apparent Polar Wander for Africa (MPW, Besse and Courtillot, 1991) in coincidence with the apparent wander of the hotspot reference frame (HPW as defined in Besse and Courtillot, 1991 and recalculated from Morgan, 1983) for each period of interest. This correction, which was done here assuming an equatorial rotation axis, substitutes the geographic pole of the mantle reference system for the paleomagnetic pole (i.e. axis of rotation of the Earth). For example, after correction, the 70 Ma Master Apparent Pole for Africa will be brought back to the mantle north pole, when considered in the 70 Ma mantle configuration. Thus corrected, the V1 (49°S, 102°E; paleolatitude = 6°N), V2 (17°S, 132°E; paleolatitude = 6°N) and V3 (73°N, 231°E; paleolatitude = 7°N) poles can be used to define solely the movement of the nappe. When they are relocated in their Late Cretaceous position in the hotspot reference frame as before, a corrected Euler pole of rotation and associated radius of the circular track of rotation can then be calculated (Fig. 7; Table 3). Because the corrections are quite small and considering paleomagnetic uncertainties, it is expected that this Eulerian

rotation is a good approximation of the actual movement of the nappe. The pole of rotation thus defined is much closer to the ophiolitic nappe, and implies a limited velocity of the nappe (slightly less than 2 cm/yr) in spite of the large angle of rotation (150°). The paleoposition of the ridge can then be restored to approximately 45°E in paleolongitude and 10°N in paleolatitude (Fig. 8). Of course, there is now a good agreement (within 1°) between the paleomagnetic paleolatitudes and the reconstruction. This new determination is within the uncertainty limit of the previous calculation. Therefore, in the Oman context, the TPW correction is not essential, in part because the TPW influence was fairly small between 100 and 70 Ma, but in the main because the PEP was located close to the moving block. With different geometrical relations, this correction may be inevitable in order to correctly define the paleoposition. In such (more general) cases, TPW correction could be made using a more sophisticated method which would assume that the relative movement between the rotation axis and the mantle can be described by successive Euler rotations. Nevertheless, it is not obvious that the practical use of such a method will not be limited by the uncertainties about the data and the PEP determinations.

5. Conclusions and discussion

Paleoreconstructions of continental blocks are largely based on paleomagnetic data. These data allow the determination of the paleomagnetic latitude which is equal to the mantle paleogeographic latitude in the absence of TPW. However, the paleolongitude cannot be determined. As a result the motion of any given crustal block cannot be described in its totality by paleomagnetic data alone. We show in this paper that it is possible to overcome this limitation by combining plate tectonics concepts, a mantle-anchored hotspot reference frame and paleomagnetic data. Thus, we can describe the movement of the Oman ophiolite across the Earth's surface in its totality and consequently determine both the initial paleolatitude and paleolongitude of this accretionnary plate boundary in the hotspot reference frame.

This approach is potentially quite general. Practically, it is a precise and rigorous method only when several conditions are fulfilled:

(1) Motions that truly lie along a small circle.

(2) A precise positioning of the PEP. This is the case here because of the large amplitude of the Euler rotation, the large distance between the Euler and paleomagnetic poles, and the paleo-

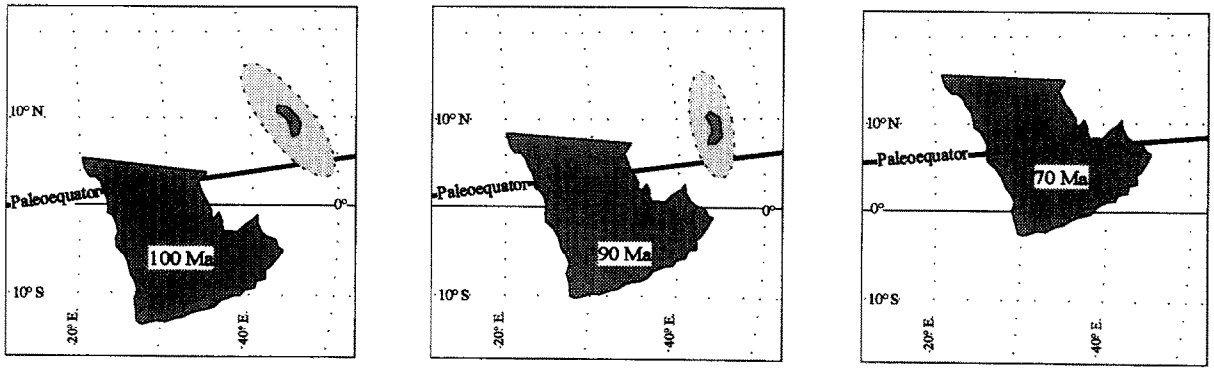


Fig. 8. Reconstructions of the position of the Oman nappe and the Arabian peninsula from 100 to 70 Ma in the hotspot reference frame. The rotation parameters used here to place the Arabian platform are those defined by Morgan (1983). The thick line is the trace of the paleomagnetic paleoequator in the hotspot reference frame. The shaded area represents the error on the paleolocation of the Oman complex. Hammer projection.

magnetic control over the full time span of movement.

(3) An independently determined TPW for the period that the crustal block was moving. This allows us to develop our calculations on the basis of the apparent wander of the mantle paleogeographic pole rather than the paleomagnetic pole. If the global paleomagnetic pole (i.e. rotation axis of the Earth) would have remained stationary, but not coincident, with respect to the mantle paleogeographic pole, between the V1 and the V3 emplacement, the TPW correction would not have modified the calculated Euler pole. The radius of the small circle fitted to the poles would be modified, however. In the current more general case, the relative position of the global paleomagnetic and the geographic poles in the hotspot reference frame would vary during rotation of the block. A TPW correction is then necessary for proper determination of the Euler pole even if, as pointed out above, several complications arise when TPW correction is large.

(4) The position of the crustal block in the hotspot reference frame at the end of its movement is known. For the Oman ophiolite, this position is known because the V3 paleomagnetic pole coincides with other African paleomagnetic poles of similar ages and because the drift of Africa with respect to the hotspot frame is known (Morgan, 1983).

With these basic conditions being fulfilled, there remain, however, some uncertainties in the specific case of the Oman ophiolite. Firstly, the question arises wether the proposed rotation is a realistic description of the actual movement of this crustal block, or represents only a rotation equivalent to a much more complex displacement. Given that only three distinct paleomagnetic poles are available, the paleomagnetic data do not allow to check the basis hypothesis of a single Euler rotation. However, as pointed out above, the circular distribution of individual VGPs from both the V1 and the V2 units provides good evidence that the actual movement was that of a single rotation during at least the V1, V2 extrusion episodes. A further uncertainty relates to the no better than approximate determination of the time of remagnetization of the V1 and the V2 units. The TPW correction therefore may be slightly incorrect. The resulting uncertainty in the position of the Euler pole should be limited to a few degrees (Fig. 7). There are also some uncertainties related to the TPW curve (Livermore et al., 1984; Andrews, 1985; Besse and Courtillot, 1991). However, the difference between the curves proposed in these three studies is very small for the period of interest (here 100 to 70 Ma). Finally, there are some uncertainties about the reconstruction of the motions of continents in the hotspot frame. However, the motion of the African plate over hotspots is rather well constrained and the error should be small.

In conclusion, this study confirmed one of Thomas (1991) hypothesis that the Oman ophiolite has undergone a large clockwise rotation, on the order of 145–150°, during its movement on the sea floor, between the detachment of the nappe and the end of the obduction. The pole of rotation was located near the present northern end of the nappe. The major contribution of this study is the determination of the paleoposition of the spreading ridge at ca. 100 Ma with a paleolatitude around 10°N and a paleolongitude around 45°E, as well as the linear velocity of the nappe between less than 1 and 5 cm/yr. We can also estimate the initial orientation of the ridge axis to be NNW–SSE to N–S. This estimate is in good agreement with the present orientation of the dike complex, inferred to be parallel to the ridge axis, which varies between N310° and north–south (Boudier et al., 1988),

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